Hide menu

Qubit and Physics

- Video: Qubit Introduction 46 sec
- Video: Superposition and Interference 15 min
- Video: Entanglement 9 min
- Video: Qubit State 10 min
- Video: Braket 8 min
- Video: Multi Qubit 5 min
- Reading: Conclusion to the Course 'Mathematical Foundations and Quantum Mechanics Essentials' 10 min
- Graded Assignment: Full Course Assessment Submitted

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Full Course Assessment

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Assignment details

Due Dec 16, 11:59 PM IST
Attempts Unlimited
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Submitted Dec 9, 11:11 PM IST
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1. In the context of quantum computing, what is significance of the identity matrix? **1 / 1 point**

- (•) It acts as the neutral element in matrix multiplication, meaning any matrix multiplied by it remains unchanged.
- () It is only applicable in classical computing, not in quantum computing.
- () It represents a matrix with all entries equal to zero.
- () It is used to calculate the determinant of other matrices.

Correct
Correct! The identity matrix is the neutral element in matrix multiplication.

2. Which property is unique to Hermitian matrices in quantum computations? **1 / 1 point**

- () Their rows are always orthogonal.
- () They are always invertible.
- (•) Their eigenvalues are always real.
- () They are equivalent to their transpose.

Correct
Correct! Hermitian matrices have real eigenvalues, which is a significant property in quantum mechanics.

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3. What is the result of multiplying the complex numbers **1 / 1 point**

$$(2+3i)$$

and

$$(1-4i)$$

in the context of quantum computing?

- () $11+7i$
- () $5-10i$
- () $-10+5i$
- (•) $14-5i$

Correct
Correct! The multiplication is done component-wise and then combined.

4. Which operations can be performed on matrices in the context of quantum computing? **1 / 1 point**

- [x] Subtraction

Correct
Correct! Subtraction is a basic matrix operation.

- [] Integration
- [x] Addition

Correct
Correct! Addition is a basic matrix operation.

- [x] Scalar Multiplication

Correct
Correct! Scalar multiplication is a basic matrix operation.

- [] Differentiation

5. What is the impact of quantum computing on solving complex problems compared to classical computing? **1 / 1 point**

- (•) Quantum computing can solve certain complex problems exponentially faster than classical computing.
- () Quantum computing is slower than classical computing for all types of problems.
- () Quantum computing solves problems at the same speed as classical computing.
- () Quantum computing has no impact on problem-solving speeds.

Correct
Correct! Quantum computing uses quantum bits (qubits) which can represent multiple states simultaneously, significantly speeding up computations.

6. What is the result of dividing the complex number **1 / 1 point**

$$(6+2i)$$

by

$$(3-i)$$

in quantum computations?

- (•) $2+i$
- () $3-2i$
- () $2-i$
- () $1+3i$

Correct
Correct! Dividing

$$(6+2i)$$

by

$$(3-i)$$

using the complex conjugate method results in

$$2+i$$

.

7. Which of the following operations are characteristics of classical computers but not quantum computers? **1 / 1 point**

- [] Data processing using superposition
- [] Simultaneous evaluation of multiple states
- [x] Data representation using binary bits

Correct
Correct! Classical computers use binary bits for data representation.

- [x] Operations based on deterministic algorithms

Correct
Correct! Classical computers often rely on deterministic algorithms.

8. When performing matrix multiplication in quantum computing, what is the requirement for the dimensions of the two matrices? **1 / 1 point**

- (•) The number of columns in the first matrix must equal the number of rows in the second matrix.
- () Both matrices must have the same number of columns.
- () The number of rows in the first matrix must equal the number of columns in the second matrix.
- () Both matrices must have the same number of rows.

Correct
Correct! This is a fundamental requirement for matrix multiplication.

9. What is the significance of eigenvalues and eigenvectors in quantum computations? **1 / 1 point**

- () They are used in the calculation of tensor products.
- (•) They help in determining the probabilities of different quantum states.
- () They help in performing scalar multiplication of matrices.
- () They are used to add and subtract complex numbers.

Correct
Correct! Eigenvalues and eigenvectors play a crucial role in understanding quantum states and their probabilities.

10. Which concepts are fundamental to understanding probability in quantum computing? **1 / 1 point**

- [] Matrix inversion
- [x] Mutually exclusive events

Correct
Correct! Mutually exclusive events are fundamental in probability theory.

- [] Transcendental numbers
- [x] Independent events

Correct
Correct! Independent events are also fundamental in understanding probability.

11. Which of the following are properties of unitary matrices in the context of quantum computing? **0.8 / 1 point**

- [] All entries in a unitary matrix are real numbers.
- [] The matrix is equal to its conjugate transpose.
- [] The determinant of a unitary matrix is always 1.
- [] The matrix is equal to its inverse.
- [x] All eigenvalues have an absolute value of 1.

Correct
Correct! Unitary matrices have eigenvalues with an absolute value of 1.

You didn't select all the correct answers

12. Which property is unique to unitary matrices in quantum computations? **1 / 1 point**

- (•) They preserve the inner product of vectors
- () They are always symmetric
- () Their determinant is always zero
- () They have only real eigenvalues

Correct
Correct! Unitary matrices preserve the inner product of vectors, which is essential in quantum computing.

13. Which of the following principles are essential for understanding quantum computing? **0.8333333333333334 / 1 point**

- [x] Superposition

Correct
Correct! Superposition allows qubits to exist in multiple states simultaneously, a fundamental principle of quantum computing.

- [x] Interference

Correct
Correct! Interference is used in quantum algorithms to amplify correct solutions and cancel out incorrect ones.

- [] Wave function collapse
- [] Classical bits
- [x] Entanglement

Correct
Correct! Entanglement enables qubits to be interconnected, which is crucial for quantum algorithms and protocols.

- [] Error correction

You didn't select all the correct answers

14. What principle of quantum mechanics explains why measuring one particle of an entangled pair immediately determines the state of the other particle? **1 / 1 point**

- () Quantum interference
- () Quantum superposition
- (•) Quantum entanglement
- () Wave function collapse

Correct
Correct! Quantum entanglement is the phenomenon where the quantum states of two or more objects are interconnected, such that the state of one instantly influences the state of the other, no matter the distance separating them.

15. Which statements are true about the representation of qubit states in quantum mechanics? **1 / 1 point**

- [] Qubit states are represented as classical bits
- [] Qubit states are always in a definite state of 0 or 1
- [x] Qubit states can be in a superposition of 0 and 1

Correct
Correct! Qubit states can exist in a superposition of both 0 and 1 simultaneously.

- [] Qubit states cannot be entangled
- [x] Qubit states can be represented using column vectors

Correct
Correct! Qubit states are often represented using column vectors in quantum mechanics.

16. In quantum computation, why is the concept of superposition important? **1 / 1 point**

- () It prevents quantum states from collapsing during computation.
- () It ensures that quantum entanglement can occur between qubits.
- (•) It allows qubits to represent both 0 and 1 simultaneously, increasing computational power exponentially.
- () It provides a way to measure quantum interference patterns.

Correct
Correct! Superposition enables qubits to perform many calculations at once, significantly enhancing the computational capabilities of quantum computers.

17. In Bra-Ket notation, what does the Ket $|\psi\rangle$ represent? **1 / 1 point**

- () A row vector representing a quantum state.
- (•) A column vector representing a quantum state.
- () The inner product of two quantum states.
- () An operator acting on quantum states.

Correct
Correct! The Ket $|\psi\rangle$ represents a column vector in Hilbert space.

18. Which of the following statements correctly describe the concept of superposition in quantum mechanics? **1 / 1 point**

- [x] A quantum system can exist in multiple states simultaneously.

Correct
Correct! Superposition means that a quantum system can exist in a combination of multiple states simultaneously.

- [] Superposition only applies to systems with two quantum states.
- [x] Measurement causes the superposition to collapse to one of the possible states.

Correct
Correct! Measurement causes the superposition state to collapse into one of the possible states.

- [x] The probabilities of the outcomes are determined by the probability amplitudes.

Correct
Correct! The probabilities of different outcomes in a superposition are determined by the probability amplitudes.

- [] Superposition is a result of the entanglement of quantum states.

19. What are the key characteristics of quantum bits (qubits)? **1 / 1 point**

- [x] Measurement outcomes are probabilistic.

Correct
Correct! Measurement outcomes for qubits are probabilistic due to their quantum nature.

- [] They follow the principles of classical mechanics.
- [x] They can be entangled with other qubits.

Correct
Yes! Entanglement is a vital property of qubits.

- [x] They can exist in multiple states simultaneously.

Correct
Correct! This is known as superposition.

- [] They can be stored and retrieved from classical memory.

20. Which of the following best represents a qubit in a superposition state? **1 / 1 point**

- () 0
- (•) $\frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)$
- () $|0\rangle$
- () $\frac{1}{2}(|0\rangle + |1\rangle)$

Correct
Correct! This is a common representation of a qubit in a superposition state, showing equal probability amplitudes for $|0\rangle$ and $|1\rangle$.